

Name: Shupthah
Neelakandan

Roll no: AM-PS-ISMAT 25015

Physics

Assignment - 1

classmate

Date
Page

Chapter 1 - Physical Quantities

1. a) Singapore to New York, USA distance = 15339 km
[1 km = 1000 m]

$$\text{distance} = 15339 \times 10^3 \text{ m}$$

$$\text{distance with 3 s.f} = 1.53 \times 10^7 \text{ m}$$

b) distance = 15339 km

$$= 15339 \times 10^3 \text{ m}$$

$$= 15.3 \times 10^6 \text{ m}$$

$$[10^6 \text{ m} = 1 \text{ Mm}]$$

$$= 15.3 \text{ Mm}$$

2. $4.11 \text{ nm} = 4.11 \times 10^{-9} \text{ m}$

$$[10^{-9} \text{ m} = 1 \text{ nm}]$$

$$3.24 \mu\text{m} = 3.24 \times 10^{-6} \text{ m}$$

$$[10^{-6} \text{ m} = 1 \mu\text{m}]$$

$$73.2 \text{ Gm} = 73.2 \times 10^{10} \text{ m}$$

$$[10^{10} \text{ m} = 1 \text{ Gm}]$$

3. side of cube = 3.2 cm (2 s.f)

$$[1 \text{ cm} = \frac{1 \text{ m}}{100}]$$

$$\text{side of cube in metre} = 3.2 \times 10^{-2} \text{ m}$$

$$\text{Volume of the cube} = (3.2 \times 10^{-2})^3$$

$$= 32.768 \times 10^{-6}$$

$$= 3.2768 \times 10^{-5}$$

$$= 3.3 \times 10^{-5} \text{ m}^3 (2 \text{ s.f})$$

$$\begin{array}{r} 32 \quad 1024 \\ 32 \quad 32 \\ \hline 64 \quad 2048 \\ 96 \quad 3072 \\ \hline 1024 \quad 32768 \end{array}$$

4. Density of lead = 11.3 g/cm^3 (3 s.f)

$$\left[\begin{array}{l} 1 \text{ Kg} = 1000 \text{ g} \quad 1 \text{ g} = 10^{-3} \text{ Kg} \\ 1 \text{ cm} = 10^{-2} \text{ m} \end{array} \right]$$

Density of lead in Kg/m^3

$$\Rightarrow 11.3 \text{ g/cm}^3 \Rightarrow 11.3 \times 10^{-3}$$

$$\Rightarrow 11.3 \times 10^{-3} \times 10^6 \text{ Kg/m}^3$$

$$\Rightarrow 11.3 \times 10^3 \text{ Kg/m}^3$$

$$\Rightarrow 1.13 \times 10^4 \text{ Kg/m}^3 \text{ (3 s.f)}$$

5. 1 billion seconds = 10^9 s

$$\begin{aligned} \text{In 365 day year no of seconds} &= (24 \times 3600) \times 365 \\ &= 365 \times 86400 \\ &= 31536000 \text{ s/year} \end{aligned}$$

no. of years old when 1 billion seconds from now

$$\Rightarrow \frac{10^9 \text{ s}}{31536000 \text{ s/year}}$$

$$\Rightarrow \frac{10^6}{31536} \text{ years} \Rightarrow 31.708 \text{ years}$$

$$\Rightarrow 31.7 \text{ years (3 s.f)}$$

$$\begin{array}{r} 31.708 \\ 31536 \overline{) 1000000000} \\ \underline{946080} \\ 539200 \\ \underline{315360} \\ 2233400 \\ \underline{2207520} \\ 258800 \\ \underline{2522880} \end{array}$$

6. Critical mass of neptunium = $60 \text{ kg} = 60 \times 10^3 \text{ g}$
 $[1 \text{ kg} = 1000 \text{ g}]$

Density of neptunium element = 19.5 g/cm^3

$\left[\text{Density} = \frac{\text{mass}}{\text{Volume}} \right]$

Volume of neptunium = $\frac{\text{mass}}{\text{density } \rho} = \frac{60 \times 10^3 \text{ g}}{19.5 \text{ g/cm}^3}$

$$= \frac{600 \times 10^3 \text{ cm}^3}{195}$$

$$= 3.07692 \times 10^3 \text{ cm}^3$$

$$= 3076.92 \text{ cm}^3$$

Volume of sphere = $\frac{4}{3} \pi r^3$

$$3076.92 = \frac{4}{3} \pi r^3$$

$$3 \times 3076.92 = r^3$$

$$4 \times 3.14$$

$$r = \sqrt[3]{\frac{9230.76}{12.56}}$$

$$= \sqrt[3]{\frac{2307.69}{3.14}}$$

$$= \sqrt[3]{734.9}$$

$$\approx 9.02$$

radius of neptunium sphere $\approx 9 \text{ cm}$

$$7. \quad F = \frac{G m_1 m_2}{r^2}$$

$F \rightarrow$ Gravitational force b/w two planets
 $m_1, m_2 \rightarrow$ mass of two planets
 $r \rightarrow$ distance between two planets

$$\left[\begin{array}{l} F = ma \quad m \rightarrow \text{SI unit} = \text{kg} \\ a \rightarrow \text{SI unit} = \text{m/s}^2 \\ \therefore F \rightarrow \text{SI unit} \quad \text{kg m/s}^2 \end{array} \right] \text{--- (1)}$$

$$\left[m_1, m_2 \rightarrow \text{SI unit} = \text{kg}^2 \right] \text{--- (2)}$$

$$\left[r^2 \rightarrow \text{SI unit} = \text{m}^2 \right] \text{--- (3)}$$

$$F = \frac{G m_1 m_2}{r^2} \Rightarrow G = \frac{F r^2}{m_1 m_2}$$

$\parallel$
 from (1), (2), (3)

SI unit of Gravitational constant G

$$\Rightarrow \frac{\text{kg m/s}^2 \cdot \text{m}^2}{\text{kg}^2}$$

$$\Rightarrow \frac{\text{m}^3}{\text{kg s}^2}$$

$$\therefore \text{SI unit } G \rightarrow \text{m}^3/\text{kg s}^2$$

$$8. \quad C = \frac{F}{\rho v^2 A} \quad F \rightarrow \text{force} \rightarrow \text{SI unit} \rightarrow \text{kg m/s}^2$$

$$\rho \rightarrow \text{density} \rightarrow \text{SI unit} \rightarrow \text{kg/m}^3$$

$$\frac{\text{mass}}{\text{Volume}} = \frac{\text{kg}}{\text{m}^3}$$

C SI unit

$$\Rightarrow \frac{\text{kg m/s}^2}{\text{kg/m}^3 \cdot \text{m}^2/\text{s}^2 \cdot \text{m}^2}$$

$$\Rightarrow \frac{\text{m}^3 \text{s}^{-2}}{\text{m}^3 \text{s}^{-2}} = 1$$

C has no dimension as SI unit of F & $\rho v^2 A$ cancelled each other out

9. $F = m^x v^y r^z$

$m \rightarrow \text{mass}$ $v \Rightarrow \text{speed}$
 $r \rightarrow \text{radius}$

F - SI unit - kg m/s^2

Dimension $\rightarrow [M][L][T]^{-2}$

$$F = m^x v^y r^z$$

$$[M]^x [L]^y [T]^z = ([M])^x ([L])^y ([T])^z$$

$$= [M]^x [L]^y [T]^{-y} [L]^z$$

$$x = 1 \quad y = 2 \quad z = -1$$

$$\cancel{[M]^1} \cancel{[L]^1} \cancel{[T]^{-2}} = \cancel{[M]^1} [L]^2 [T]^{-2} [L]^{-1}$$

$$[L]^T = [L]^T$$

$$\therefore x = 1 \quad y = 2 \quad z = -1$$

10. No of passengers = 37203978
with 3 s.f = 3.72×10^7

11. Product of 5.3×10^6 & 8.861×10^{-6}

$\Rightarrow (5.3 \times 10^6) \times (8.861 \times 10^{-6})$

$\Rightarrow 5.3 \times 10^6 \times 8.861 \times 10^{-6}$

$\Rightarrow 46.7633$

$\Rightarrow 47$ (2 s.f.)

$$\begin{array}{r} 8861 \\ \times 53 \\ \hline 25583 \\ 44305 \\ \hline 467633 \end{array}$$

$$\begin{aligned}
 12. \quad & \overset{2 \text{ sf}}{5.3} \times 10^6 - 8.861 \times 10^{-6} \\
 & = 5300000 - 0.000008861 \\
 & = 5299999.9 \\
 & = 5.29 \times 10^6 \\
 & = 5.3 \times 10^6 \quad (2 \text{ sf})
 \end{aligned}$$

$$\begin{aligned}
 13. \quad & \overset{3 \text{ sf}}{1.96} \times 10^{-4} + 7.561 \times 10^{-5} \\
 & = 0.000196 + 0.00007561 \\
 & = 0.00027161 \\
 & = 2.7161 \times 10^{-4} \\
 & = 2.72 \times 10^{-4} \quad (3 \text{ sf})
 \end{aligned}$$